STATISTICAL ANALYSIS PROJECT TITLES AND OBJECTIVES

BTECAI25301 – Applied Statistical Analysis

Group 1:
Shrikant Malpani
Soham Koli
Madhav Rathod
Dhruv Valambhiya
Pritesh Shaha

Project 1: Predictive Statistical Analysis of a Real-World Dataset Using R/Python

Objectives:

1. Clean and organize a real-world dataset.
2. Perform descriptive analysis and visualization.
3. Apply probability, correlation/regression and hypothesis-testing techniques.
4. Use R/Python and AI-assisted tools to support statistical analysis and interpretation.

Group 2:

Atharv Kulkarni
Mohammad Ziya Khan
Aifaz Khan
Yash Ambuskar
Vaibhav Sachin

Project 2: Analysis of Student Academic Performance and Factors Affecting Achievement

Objectives:

1. Analyze students' marks, attendance and study hours using descriptive statistics.
2. Examine relationships between study hours, attendance and academic performance using correlation.
3. Test whether academic performance differs between selected student groups.

Group 3:
Soham Mokashi
Azlan Mulani
Parth Kolekar
Krishna Palange
Priyanshu Gujar

Project 3: Analysis of Customer Satisfaction and Service Quality

Objectives:

1. Construct frequency distributions and graphical summaries of customer ratings.
2. Calculate descriptive measures of customer satisfaction.
3. Examine associations between service quality and customer satisfaction

Groups 4:

Anuj Ghodke
Rajbeer Sigh Sachar
Hriday Panchmukh
ruturaj Bhosale
yuvaraj bhosale

Project 4: Statistical Analysis of Employee Performance and Training

Objectives:

1. Summarize employee performance and training data.
2. Investigate the relationship between training hours and performance.
3. Test whether trained and untrained employees differ significantly in performance.

Group 5:

Sharvari Pawar
Poorva Sawant
Tanushree Dharmalingam
Nehal Chaudhari

Project 5: Analysis of Employee Job Satisfaction and Workplace Factors

Objectives:

1. Analyze job satisfaction scores using descriptive measures and box plots.
2. Examine relationships between job satisfaction, working hours and experience.
3. Compare satisfaction across departments

Group 6:

Isha Honkalse
Shreya Kalekar
Avani Ostwal
Afiya Farid Ansari

Project 6: Analysis of Sales and Customer Purchasing Patterns

Objectives:

1. Analyze purchase frequency and transaction values.
2. Identify purchasing patterns using tables and graphs.

3. Examine relationships between purchase frequency, purchase value and customer characteristics.

Group 7:
Jay Sarode
Shashwat Sinha
Sarah Shaikh
Krish Patel
Shardul Ware

Project 7: Analysis of Social Media Usage and Academic Performance

Objectives:

1. Analyze daily social-media usage using descriptive statistics.
2. Study the relationship between social-media usage and academic scores.
3. Test whether academic performance differs among different usage groups.

Group 8:

Dayanidhi Shinde
Aarya Darwatkar
Vebew
Afeef Shaikh
Sujal Sonar

Project 8: Analysis of Healthcare Data: Patient Characteristics and Health Outcomes

Objectives:

1. Summarize patient demographic and health variables.
2. Construct confidence intervals for selected population parameters.
3. Apply appropriate tests to investigate differences and associations.